# Yanan He

102 N. Chauncey Ave, West Lafayette, IN 47906

https://sites.google.com/view/yananhe/home

## EDUCATION

**Purdue University**, West Lafayette, IN — Aug 2023 – May 2026

Bachelor of Science in Computer Science & Mathematics-Statistics — GPA: 3.96/4.0

Awards: Dean's List (2023–2025), Semester Honor (2023–2025)

**Relevant Coursework:**

AI & CS: Machine Learning, Data Mining, Intro to AI, Algorithms Analysis, Data Structures, System Programming, Computer Architecture

Math & Stats: Numerical Methods, Stochastic Processes, Linear Algebra, Probability, Statistical Inference, Real Analysis

## RESEARCH INTERESTS

AI for science; trustworthy and interpretable AI; foundation models and representation learning.

## PUBLICATIONS

* Corresponding author.

**Conference Papers**

[C.1] **He, Y.**, Wen, Y., Wang, X., & Ma, T.* (2026). MTS-JEPA: Multi-Resolution Joint-Embedding Predictive Architecture for Time-Series Anomaly Prediction. *arXiv:2602.04643*. Under review at *ICML 2026*.

**Journal Articles**

[J.1] Li, J., Chang, H., Du, S., et al., **He, Y.***, & He, M.* (2025). Development and validation of a web-based machine learning model for predicting early neurological deterioration following stroke thrombolysis: A multicenter study. *Journal of Medical Internet Research*, 27, e77858. 10.2196/77858.

[J.2] Li, J., Zhang, C., Zhang, H., Gu, X., **He, Y.***, & He, M.* (2025). Socioeconomic gradients in inflammatory risks and global stroke mortality. Under review at *The Lancet Neurology.*

[J.3] He, M., Zhang, S., Liu, X., **He, Y.**, Gu, X.*, & Yue, C.* (2025). Global, regional, and national epidemiology of ischemic stroke in young adults, 1990–2021. *Journal of Neurology*, 272(5), 354. 10.1007/s00415-025-13082-4.

## RESEARCH EXPERIENCE

**Stony Brook University**, Stony Brook, NY — May 2025 – Present

Research Assistant | Machine Learning — Advisor: Tengfei Ma

Project I: MTS-JEPA: Multi-Resolution Joint-Embedding Predictive Architecture for Time-Series Anomaly Prediction

- Led development of a self-supervised framework for time-series anomaly prediction, introducing a **multiscale JEPA-style architecture** to model long-range temporal dependencies.
- Addressed the **JEPA representation collapse** problem via a **soft-quantized codebook latent space**, improving stability of non-contrastive training and representation consistency.
- Achieved **state-of-the-art performance** on multiple benchmarks, improving **AUC and F1 by up to 5%**; manuscript submitted to **ICML 2026**.

Project II: Walking in the Ball: Hyperbolic Latent Navigation for Structured Thought Generation

- Leading a research project on hyperbolic latent navigation for **large language model reasoning**, exploring how non-Euclidean geometry can better model hierarchical reasoning structures.
- Introduced a geometric reasoning framework that embeds latent thought vectors in a **Poincaré ball** and models reasoning trajectories through **geodesic flows in hyperbolic space.**
- Designing the training framework and experimental pipeline to evaluate hyperbolic latent navigation for multi-step reasoning; target submission to **NeurIPS 2026.**

**Nanjing Medical University**, Jiangsu, China — Feb 2025 – Sep 2025

Research Assistant | AI for healthcare — Advisor: Mingli He

Project: Machine Learning and Global Epidemiological Modeling for Stroke Risk and Outcome Prediction

- Served as **co-corresponding author**, providing methodological oversight and analytical guidance to an interdisciplinary research team.
- Contributed to the **design and evaluation** of an interpretable XGBoost model for disease progression prediction and clinical decision support (ROC–AUC > 0.90).

- Guided **statistical and machine learning modeling**, applying generalized additive models (GAMs) to study nonlinear socioeconomic–risk interactions in GBD-based stroke research.
- Contributed to **manuscript preparation** and provided continued technical input during peer review and revision.

**Fudan University**, Shanghai, China — Nov 2024 – Apr 2025
Research Assistant | Data Science — Advisor: Chaoyan Yue
Project: Global Epidemiological Modeling of Ischemic Stroke in Young Adults (1990–2021)

- Analyzed global ischemic stroke trends (1990–2021) in young adults using GBD 2021 data, employing **Joinpoint regression** and **Bayesian APC models** to forecast disease burden.
- Contributed to **data cleaning and statistical analysis**, and assisted with figure preparation and drafting parts of the methods and results sections to ensure clear and consistent data presentation.

## SKILLS

**Programming Languages:** Python, Java, C/C++, Assembly, R, SQL, MATLAB, JavaScript, HTML/CSS
**Expertise:** Deep Learning, Machine Learning, Statistical Modeling, PyTorch, TensorFlow, scikit-learn, Transformers
**DevOps & Cloud:** Git, Kaggle, Linux, Docker, AWS (EC2, S3, Lambda), Google Cloud, Vast.ai
**Web Development:** Flask, FastAPI, React.js, Node.js, RESTful API design, PostgreSQL, JSON
**Languages:** Chinese (Native), English (Fluent)